

Radio LAB 2: Characterizing a Radio Telescope: Noise and Temperature Calibration

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1. INTRODUCTION

As with any observational method in astronomy, there are sources of noise in radio telescopes which can affect observations. The system temperature determines fluctuations for spectral values. The temperature of the whole system includes noise from electronics, astronomical sources, background temperature, atmospheric contribution, and more.

The SRT has a noise diode which allows us to measure the system temperature. However, this diode can deteriorate over time and therefore, give an inaccurate estimate. So, it is needed to check the calibration performance by comparing a calculated value of the system temperature with the SRT-reported value. We can also compare our spectral-line observations with a published and validated survey.

2. OBSERVATIONS

In order to collect data for this lab, I prepared an observing script to submit to the Sterling Hall rooftop SRT. I set the frequency to be 1420.4 MHz, which is the frequency of the 21-cm line of atomic hydrogen and set the mode to 4, which returns 156 frequency bins. Then, I chose a position in the outer galaxy of the Milky Way to record observations. I selected a galactic longitude of 110° and latitude of 0°. I applied an elevation offset of 30° to run the calibration diode at a position off the galactic plane. I returned to the initial position on the galactic plane and integrated for 15 minutes. A copy of the observing script can be seen below (Figure 1). Observations were collected on November 5, 2025. The average elevation of these observations is 58°, excluding the elevation offset for the calibration diode. The system temperature as reported by the SRT is 2412K.

```
: record radio2_data_lauren.rad
: freq 1420.4 4
: galactic 110 0
: offset 0 30
: noisecal
: offset 0 0
:900
: roff
```

Figure 1. SRT observing script

In addition to observations from the Sterling SRT, I will also be using a spectral line data cube from the Leiden-Dwingeloo Survey (LDS) – a professional HI survey.

3. DATA ANALYSIS

Note: The Python code used for analysis is shown at the end in the Appendix (Section 5).

3.1. Data Preparation

I began by exploring the raw data file in Visual Studio to better understand the file structure. I read the file into Python while ignoring lines beginning with ‘#’ (as these denote comments) and labeling columns corresponding to the file format detailed in the UW SRT Manual. Looking at the time stamps in the raw file and Python dataframe, I separated the data into calibration data – which was recorded during the elevation offset – and observing data – which was recorded after the 900 second integration time began. This observing data (2) will be used for further analysis.

	time	az	el	az_offset	el_offset	freq_1_bin	channel_width	spectrometer_mode	num_channels	freq_channel_1	...
8	2025:308:22:56:22	41.9	57.0	0.0	0.0	1419.79	0.007812	4	156	30.6	...
9	2025:308:22:56:30	41.9	57.0	0.0	0.0	1419.79	0.007812	4	156	78.0	...
10	2025:308:22:56:38	41.9	57.0	0.0	0.0	1419.79	0.007812	4	156	36.7	...
11	2025:308:22:56:47	41.9	57.1	0.0	0.0	1419.79	0.007812	4	156	-20.9	...
12	2025:308:22:56:55	41.9	57.1	0.0	0.0	1419.79	0.007812	4	156	38.7	...
13	2025:308:22:57:02	41.9	57.1	0.0	0.0	1419.79	0.007812	4	156	5.2	...
14	2025:308:22:57:10	41.9	57.1	0.0	0.0	1419.79	0.007812	4	156	31.5	...
15	2025:308:22:57:18	41.9	57.1	0.0	0.0	1419.79	0.007812	4	156	44.3	...
16	2025:308:22:57:26	41.9	57.1	0.0	0.0	1419.79	0.007812	4	156	9.6	...
17	2025:308:22:57:34	41.9	57.1	0.0	0.0	1419.79	0.007812	4	156	55.3	...

Figure 2. Excerpt of observing data

The dataframe contains 113 rows/observations and 165 columns, corresponding to 9 fixed fields and 156 spectral values for 156 frequency channels.

3.2. Producing Averaged HI Spectrum

In order to calculate the average HI spectrum, I took the mean across all 113 observations for each frequency channel. To calculate the frequency array, I began with the frequency bin value for the first frequency channel, then added the channel width to obtain the second frequency value, and so on, such that the i th frequency is given by

$$freq_i = freq_1 + i \cdot \Delta = 1419.79 + i \cdot 0.007812$$

where i ranges from 0 to 155 and the initial frequency and channel width values are those listed in the dataframe. We can visualize the average HI spectrum as a function of this frequency array in 3 below.

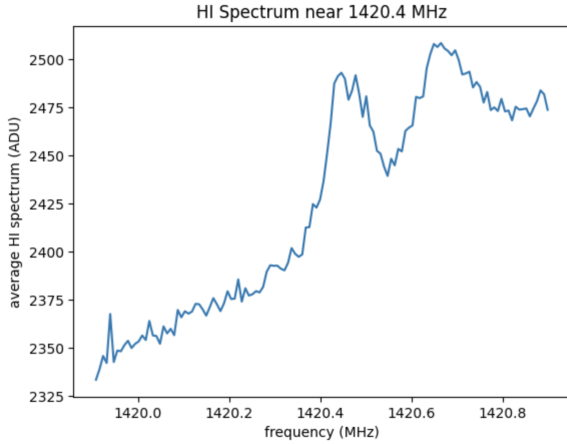


Figure 3. Average HI spectrum by frequency

We can also consider this relationship in terms of antenna temperature versus radial velocity. Using the non-relativistic Doppler shift equation (1), we can convert observed frequency into topocentric radial velocities:

$$v = \frac{c(freq_0 - freq_{obs})}{freq_{obs}} \quad (1)$$

where c is the speed of light (in km/s), $freq_0$ is the rest frequency (1420.4 MHz), and $freq_{obs}$ corresponds to the values in the aforementioned frequency array. Since the average HI spectrum is in units of ADU/counts, we can convert this to Kelvin using the gain correction constant listed in the output file. This value (0.15) represents the ratio of temperature in K per count. We can then visualize this relationship in Figure 4).

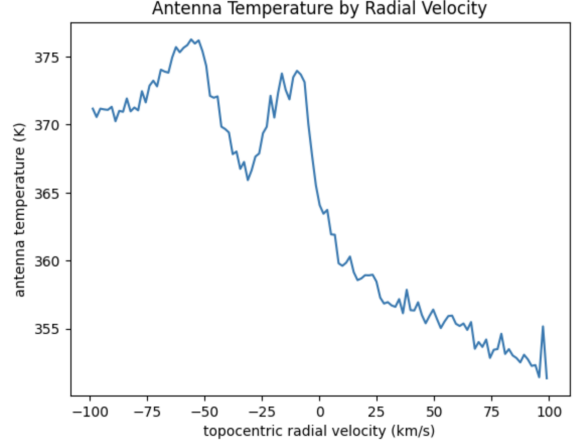


Figure 4. Antenna temperature by radial velocity

For a baseline calibration, we will use the technique of baseline fitting, which assumes that over a narrow range of frequencies, the baseline is flat. I performed a linear polynomial fit to frequency channels with no obvious HI emission to model the spectral baseline which will then be subtracted from the average HI spectrum. I considered this in terms of radial velocities, and by inspecting the plot, chose velocities with absolute magnitude between 90 and 100 km/s to model the baseline HI emission. Using this polynomial fit, I calculated the spectral baseline value for each radial velocity input. This baseline was then subtracted from the average HI spectra, which can be shown in Figure 5 as a function of topocentric radial velocities.

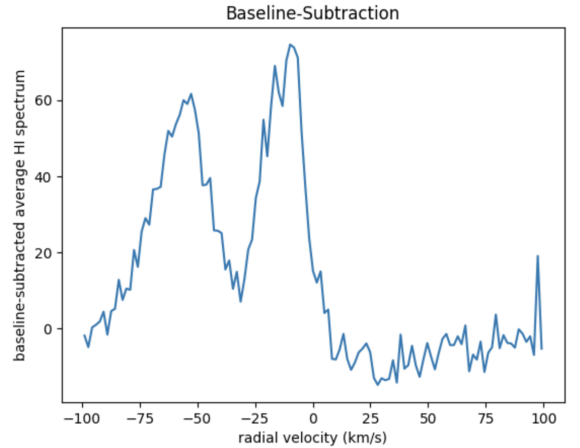


Figure 5. Baseline fitting and subtraction

3.3. Extracting the Corresponding HI Spectrum from the LDS Survey

Given my galactic coordinates of $(\ell, b) = (110^\circ, 0^\circ)$, the corresponding rectangular coordinates are $(x, y) = (220, 180)$. With these, we can extract a single spectrum from the cube at this position. I extracted velocity information from the header of the data cube which I used to the velocity in km/s. The plot below (Figure 6) shows this velocity by the brightness temperature from the extracted spectrum.

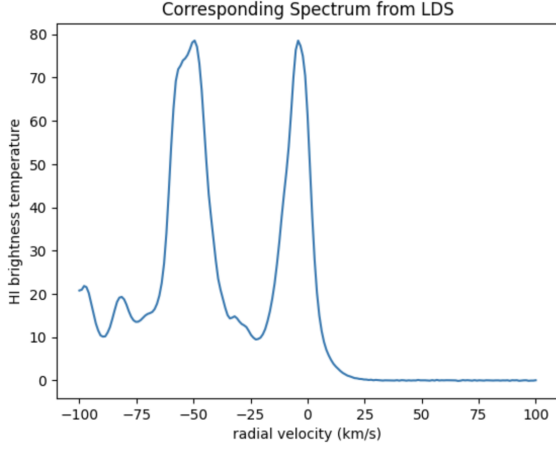


Figure 6. LDS: Radial velocity by brightness temperature

The main peaks in Figure 5 and Figure 6 seem to be closely aligned, with peaks centered near velocity of 0 km/s and -50 km/s. To find the scaling value between these curves, I first averaged the LDS cube over 12×12 subcube to account for resolution differences. I then calculated the integrated intensity for both the LDS spectrum by velocity relationship and the SRT baseline-subtracted average HI spectra values by radial velocity, while converting the spectra to Kelvin by multiplying by the correction constant. I calculated the main beam efficiency (η_{mb}) by dividing the integrated value from the LDS curve by the integrated value from the SRT curve. This resulted in a value of

$$\eta_{mb} = 0.632$$

which can be used to convert antenna temperature into brightness temperature via

$$T_B = \frac{T_A}{\eta_{mb}}.$$

The baseline-subtracted average HI spectrum can be converted into antenna temperature, then into brightness temperature. This allows us to compare the SRT and LDS curves equally. The original SRT curve (depicting antenna temperature only) is shown along with the scaled curve.

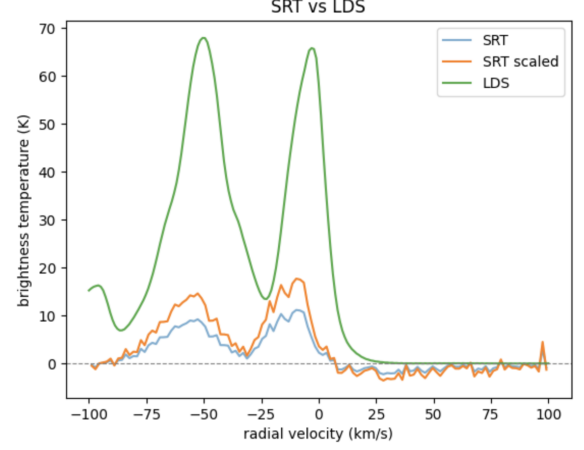


Figure 7. Spectra comparison

We see after converting all curves into brightness temperature, they have different heights. However, the spectra values appear to retain the same shape, with peaks around -50 and 0 km/s that flatten out around 25 km/s.

3.4. Calculating a New Estimate of System Temperature

We can use the radiometer equation to estimate the system temperature

$$\sigma_T = \frac{T_{sys}}{\sqrt{\Delta \nu T}}$$

where T_{sys} is the system temperature in K, T is the integration time in seconds, $\Delta \nu$ is spectral resolution in Hz, and σ_T is the noise in the spectral-line observation in K. Solving for T_{sys} , we get

$$T_{sys} = \sigma_T \cdot \sqrt{\Delta \nu T}.$$

The width of a single frequency channel is 0.0078125 MHz, or 7812.5 Hz. For frequency channels away from the HI peaks, we can measure the noise of the baseline-subtracted average HI spectrum by taking the standard deviation and converting to K. The total integration time was 900 seconds. Substituting these values and evaluating gives

$$T_{sys} = 2592.5517 \text{ K}$$

which is quite close to the reported system temperature of 2412 K.

4. CONCLUSIONS

The estimated system temperature was relatively close to what I expected, but errors still exist. Sources of random error could exist from possible interference with

118 observations or human error, which could be reduced
119 by repeating measurements. Sources of systematic error
120 could include the ambient temperature during observa-
121 tion and the noise from the electronics if not properly
122 cooled.

123 Overall, it was neat to compare data collected by us
124 to an established, professional record of similar data. I
125 think the SRT versus LDS curves differed because of the
126 difference in beam size (or from a calculation error on
127 my part).

128 5. APPENDIX

129 A. Python code is attached.